



Estd. 2000

Full Stack Development: FSD

B.Tech 1st Semester

Odd, Session 2025-26

Day-0

What is the Web?

- The **Web** = **World Wide Web (WWW)**.
It is a **system of interlinked documents (web pages)** that can be accessed via the **Internet** using a **web browser**.

Important:

- The **Internet** = A huge network of computers around the world connected via cables, satellites, and wireless signals.
- The **Web** = services and information that run **on top of the Internet**.

Components of the Web

- **Webpage**

- A document written in **HTML** (HyperText Markup Language).
- Stored on a **web server**.
- Example: about.html page of a company.

- **Web Server**

- A computer that stores and delivers webpages.
- Example: Apache, Nginx, Node.js servers.

- **Browser**

- A software (Chrome, Edge, Firefox) that requests, receives, and displays webpages.
- It understands HTML, CSS, JavaScript.

Components of the Web

- **HTTP/HTTPS** (HyperText Transfer Protocol)
 - A set of **rules** that allows browser $\leftrightarrow$ server communication.
 - Example: When you visit <https://google.com>, browser sends an HTTP request, server responds with HTML.
- **URL (Uniform Resource Locator)**
 - Address of a webpage on the web.
 - Example:
 - <https://www.amazon.in/products>
 - https $\rightarrow$ Protocol
 - www.amazon.in $\rightarrow$ Domain name (maps to server IP)
 - [/products](https://www.amazon.in/products) $\rightarrow$ Path to webpage

How the Web Works (Step-by-Step)

- You type `www.wikipedia.org` in Chrome.
- Browser converts domain name → IP address (using **DNS – Domain Name System**).
- Browser sends an **HTTP request** to the server.
- Server processes the request and sends back an **HTTP response** (usually HTML + CSS + JS files).
- Browser renders those files into a visible webpage.

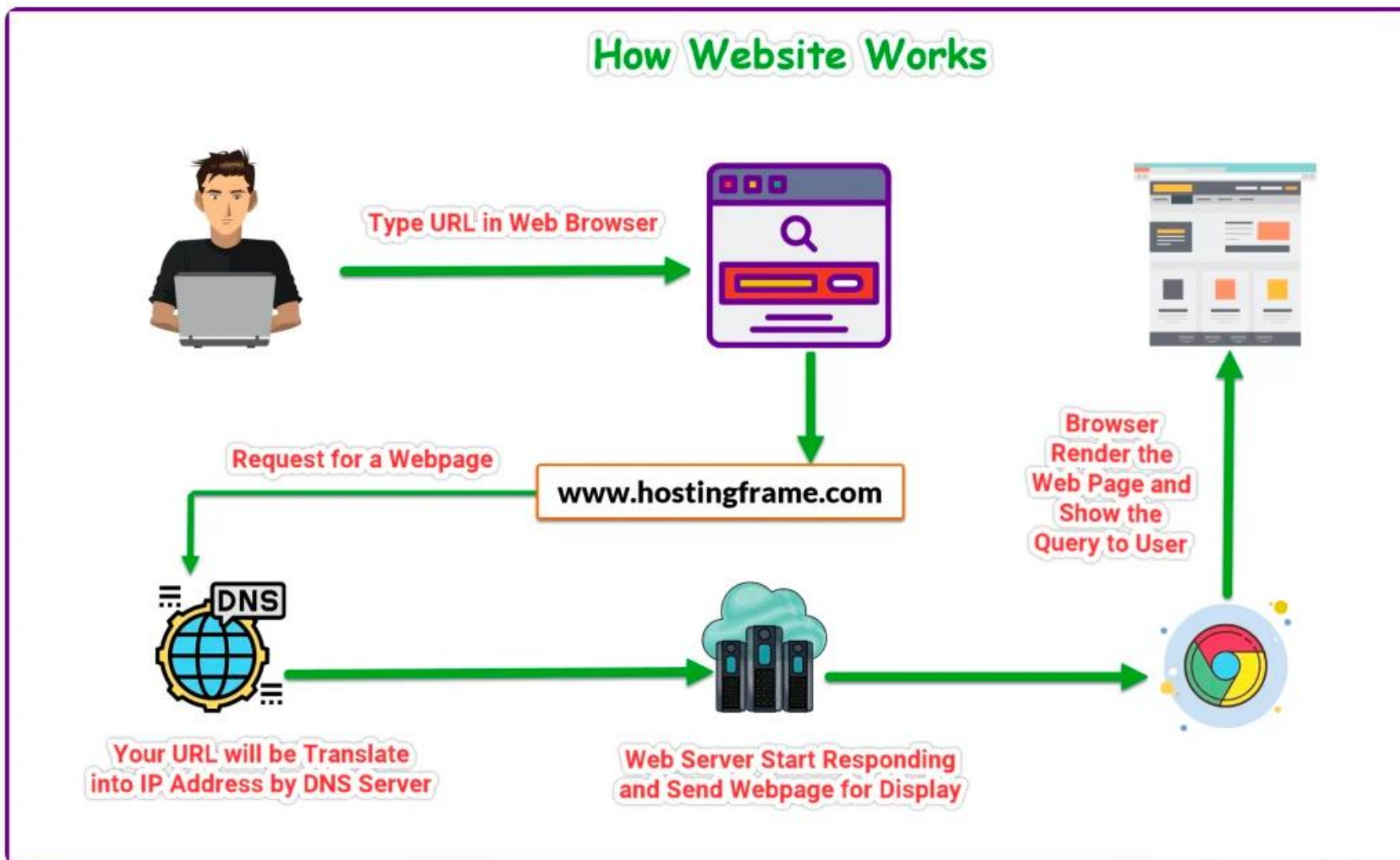
WHAT IS WEB DEVELOPMENT?

Web development refers to the process of creating websites and web applications that are accessible via the internet.

It involves a combination of different skills and technologies to design, build, and maintain websites that can range from simple static pages to complex interactive applications.

It includes aspects such as web design, web publishing, web programming, and database management. It is the creation of an application that works over the internet i.e. websites.

Basic Understanding of Websites



Basic Understanding of Websites

- When a user enters a website's URL into a web browser, the browser sends a request to the website's server.
- The server processes the request, retrieves the necessary files and data, and sends them back to the user's browser.
- The browser interprets these files (HTML, CSS, JavaScript) and displays the website's content and design.

Web Browsers

- Web browsers (like Chrome, Firefox, Safari) are applications that allow users to access information on the internet.
- Browsers render HTML, CSS, and JavaScript to display web pages.
- They also support additional features like bookmarks, extensions, and developer tools.

ABES ENGINEERING COLLEGE, GHAZIABAD

WD1 [1st Semester] | Duration: 36 Hours (3 Hrs/week)

Detailed Syllabus 1st Semester

Time stamp	TOPIC
WEEK 1 : Introduction to Web Technology, Tools & Configuration	Introduction to Web Technology
	Web Designing & Development
	Client Server Architecture
	Introduction to Text Editors
	Installation & Set up of VS Code Editor
	Introduction of Git & Git Hub
	Installation & Configuration of Git Account
WEEK 2 : Introduction to Html, Elements & Basic Tags	Introduction to HTML & HTML Features
	HTML Page Structure
	Elements in HTML(Sementic and Non Sementic)
	Types of Elements(Inline/Block)
	Text formatting: headings, paragraphs, lists, links, Images and attributes
	Comment in HTML, HTML Formatting Elements/ Special Charcter & Symbol
	Text related Elements(Quotation and Citation Elements)
	Using Multimedia (Image/video/Audio/youtube) in HTML
	Creating & Using Hyperlinks
WEEK 3 : List	HTML Lists
	Unordered Lists
	Ordered Lists
	Description Lists
	Nesting and Mixed Lists
	Lists for Navigation Menus
	Accessibility and Semantic Considerations

ABES ENGINEERING COLLEGE, GHAZIABAD	
WD1 [1st Semester] Duration: 36 Hours (3 Hrs/week)	
Time stamp	TOPIC
WEEK 4 : Table	Creating and using Tables
	Basic Table Structure
	Table Headers and Captions
	Rowspan and Colspan
	Table Attributes
	Responsive Tables
	Nested Tables
	Accessibility in Tables
WEEK 5 : Forms	The <form> Element
	Input Elements
	Label and Fieldset
	Select and Option Tags
	Textarea
	Button Element
	Form Attributes
	Input Validation
	Accessibility and Usability
WEEK 6: Introduction to Git & Local Version Control	What is Git and why use it?
	Git setup and config (git config)
	Creating a local repository: git init
	Tracking changes: add, commit, status, log
	.gitignore basics
WEEK 7 : GitHub & Remote Collaboration	Creating a GitHub account
	Connecting local repos to GitHub: remote add, push
	Cloning and pulling repositories
	GitHub repository structure and README.md
	Basic collaboration terms: fork, pull request, branch (intro)

Detailed Syllabus 1st Semester

Detailed Syllabus 1st Semester

ABES ENGINEERING COLLEGE, GHAZIABAD	
WD1 [1st Semester] Duration: 36 Hours (3 Hrs/week)	
Time stamp	TOPIC
WEEK 8: Project Version Control & Team Workflow	Branching: git branch, git checkout, git switch
	Merging branches, handling merge conflicts
	GitHub collaboration workflow
	Best practices for commits, branches, and messages
WEEK 9: CSS Basics & Box Model	CSS selectors: element, class, ID
	Color, fonts, text alignment, font size
	Box model: padding, margin, border
	Applying styles: inline, internal, external
WEEK 10: Page Layout with CSS (Float, Flexbox)	display: block, inline, inline-block
	float and clear
	Introduction to Flexbox: justify-content, align-items, flex-direction
WEEK 11: Responsive Design & Media Queries	Units: px, %, em, rem, vh, vw
	Media queries and breakpoints
	Mobile-first design
	Responsive navigation concepts
WEEK 12: Capstone Project & Portfolio Publishing	Project planning and design
	Version control in large projects
	Deploy using GitHub Pages
	Final portfolio polishing

1. Project will be assigned to each group of students.
2. The final outcome of the contest in one executable project (projects to be uploaded over GitHub)
3. Marks will be assigned through your participation and project development.

Training
Team

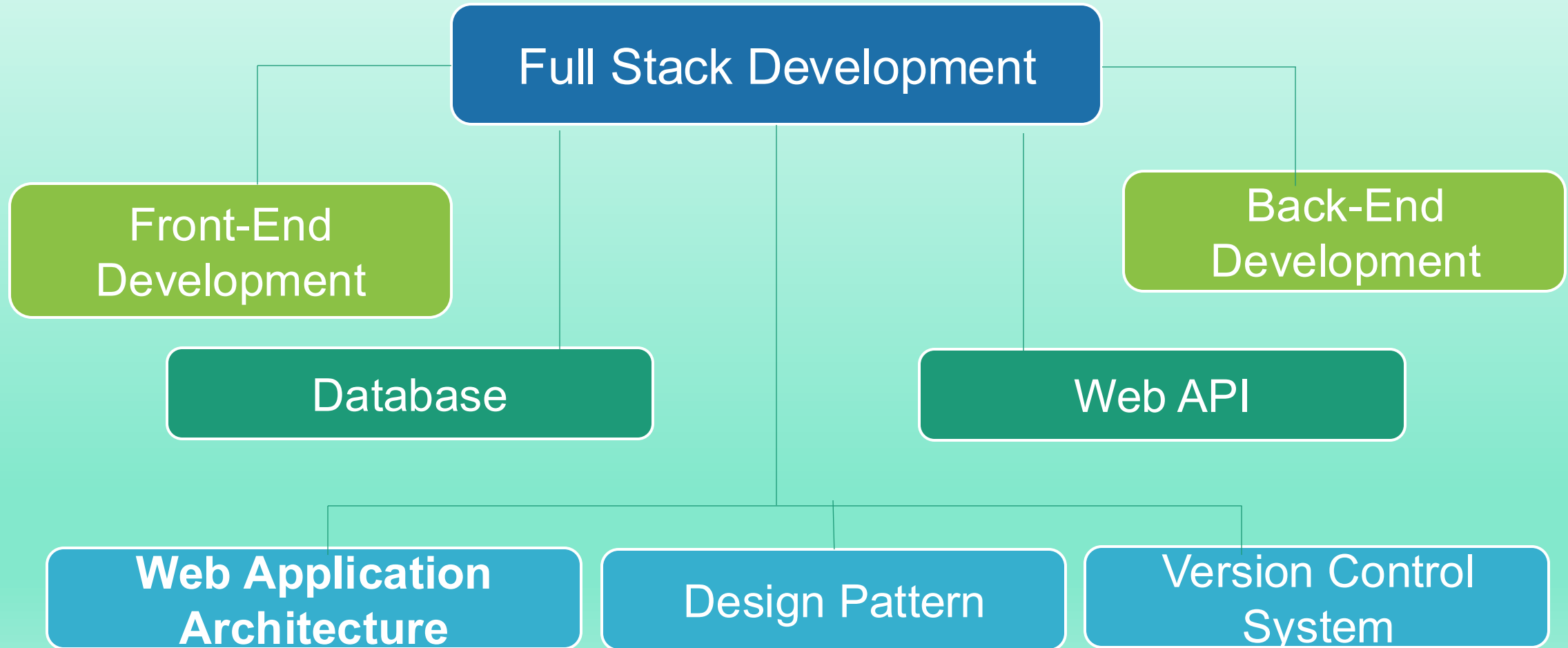


ABES ENGINEERING COLLEGE,
Full Stack Development / Training Team

2025-2026|| Odd-Even Semester

Sr. No	Name	Mobile No.	Email Id
1	Ms. Deepti Deshmukh	9284858892	deepti.deshmukh@abes.ac.in
2	Mr. Jitendra Kumar Chauhan	9971660499	jitendra.chauhan@abes.ac.in
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4	Ms. Pranshi Verma	7290912026	pranshi.verma@abes.ac.in
5	Ms. Monika	7838119776	monika@abes.ac.in
6	Mr. Rohit Pratap Singh	9528781698	rp.singh@abes.ac.in

Components of Full Stack Development



Front-End Development

- This is the client-side, the part of the application that users directly interact with. It includes the User Interface (UI) built using HTML, CSS, and JavaScript, and often a framework like React or Angular, Vue.
- **Back-End Development:**
- This is the server-side logic. It processes user requests, applies business rules, interacts with the database, and sends responses back to the front-end. **Node.js (Express), Python (Django/Flask), Java (Spring Boot), PHP (Laravel).**
- **Database**
- Databases are where application data is stored. It could be SQL (MySQL, PostgreSQL) or NoSQL (MongoDB).

API

- APIs (Application Programming Interfaces) act as bridges between front-end and back-end. REST APIs allow data to flow in a structured, standard format, usually JSON.
- An API is like a **messenger** that allows two different software applications to talk to each other and share data.
- You use a weather app. The app itself doesn't have weather data. It calls a Weather API → gets the temperature, forecast, etc. → and shows you.
- **Types** → REST API, SOAP API, GraphQL API.

REST API

- **Web Application Architecture**
- This defines how all parts of the system communicate — the client, the server, the database, and APIs. It's the blueprint. Monolithic Architecture, Client-Server Architecture, 3-Tier Architecture, **Microservices Architecture**.
- **Design Pattern:** predefined solutions/blueprints
- Design patterns are standard, reusable solutions for common problems in software development. Example: MVC (Model-View-Controller).

Full Stack Developer

Over **2 billion** Websites

Approximately **5 billion** active internet users

} **Need for enhanced
customer experience**

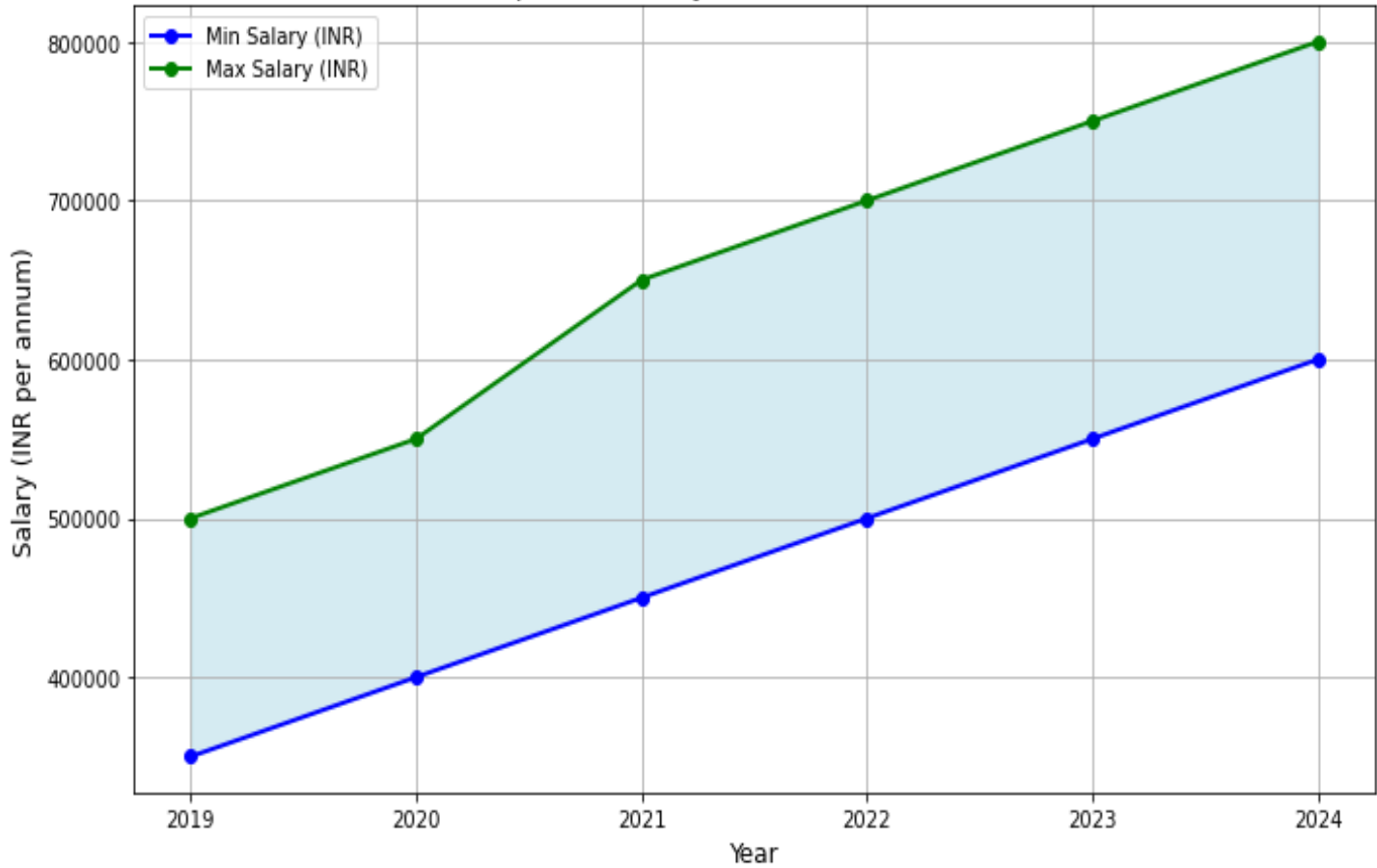
*In 2024, **Full Stack Developers** still rank as one of the top roles to be filled, often holding the **#1 or #2 spot**, closely followed by roles such as **AI/ML Engineers** and **Cloud Engineers**.*

- Among one of the most in demand job profiles for B. Tech Graduates
- Ability to work on both front-end and back-end
- Knowledge of a broad spectrum of technologies
- Easy to learn
- Provides a much wider understanding of the scope of the project
- Creative flexibility
- Handsome salary packages

The demand for full-stack developers will steadily increase in the market as new technologies emerge.

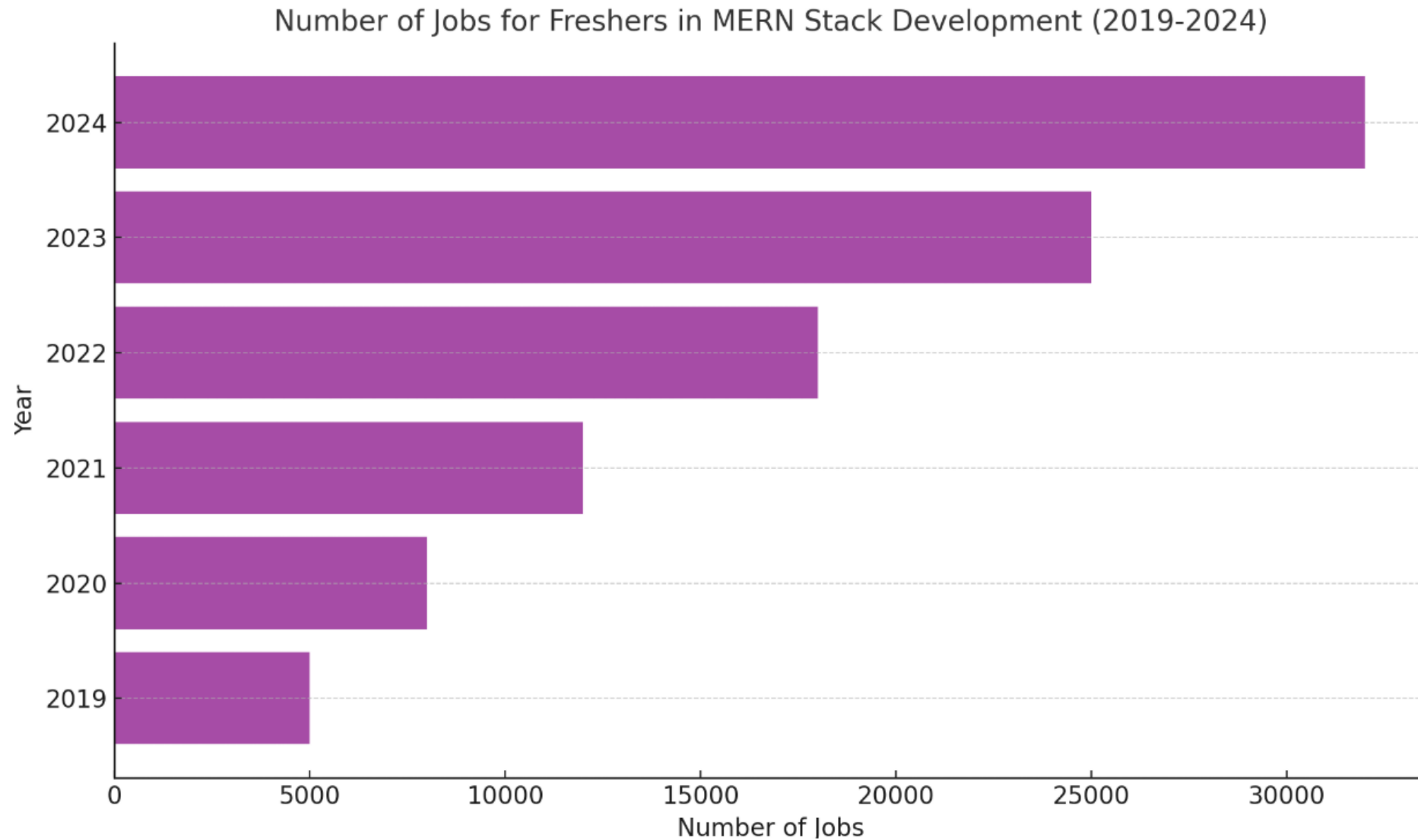
Full Stack Developer (Salary Trends)

MERN Stack Development: Salary Trends for Freshers in India (2019-2024)

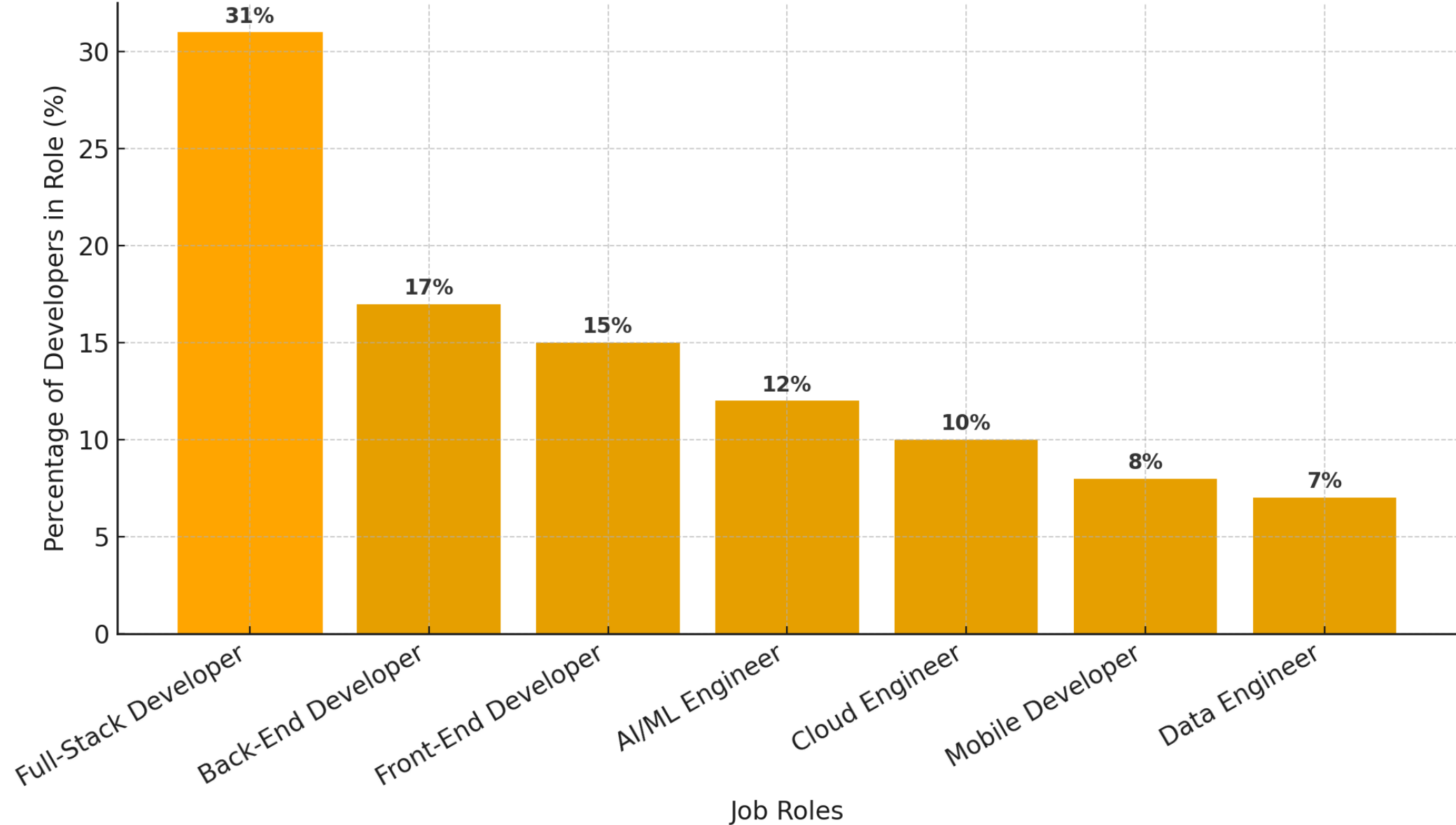


Year	Average Salary for Freshers (INR per annum)
2019	3,50,000 - 5,00,000
2020	4,00,000 - 5,50,000
2021	4,50,000 - 6,50,000
2022	5,00,000 - 7,00,000
2023	5,50,000 - 7,50,000
2024	6,00,000 - 8,00,000

Full Stack Developer (Freshers job Trends)

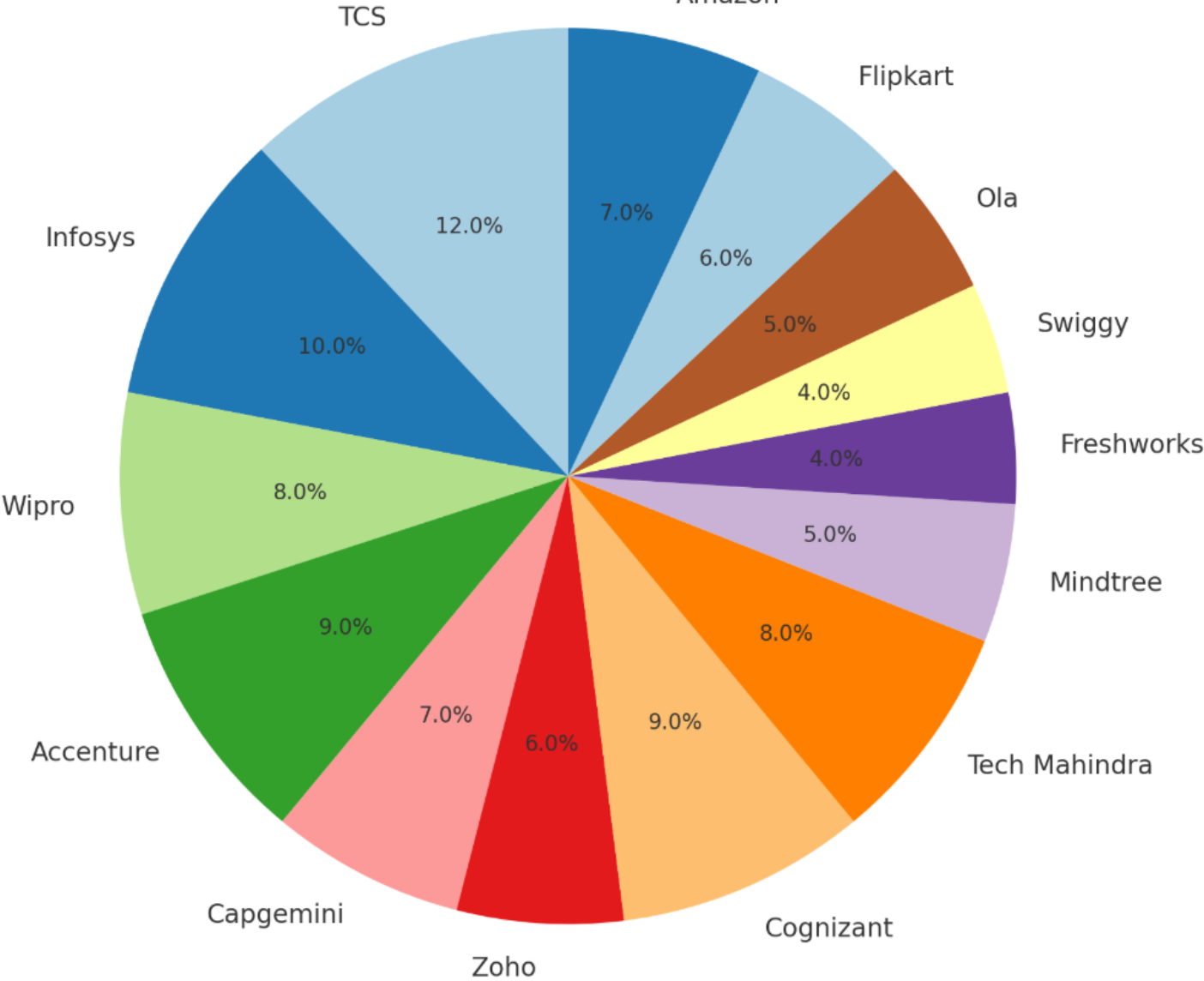


Job Role Distribution (2024-25): Full-Stack Leads

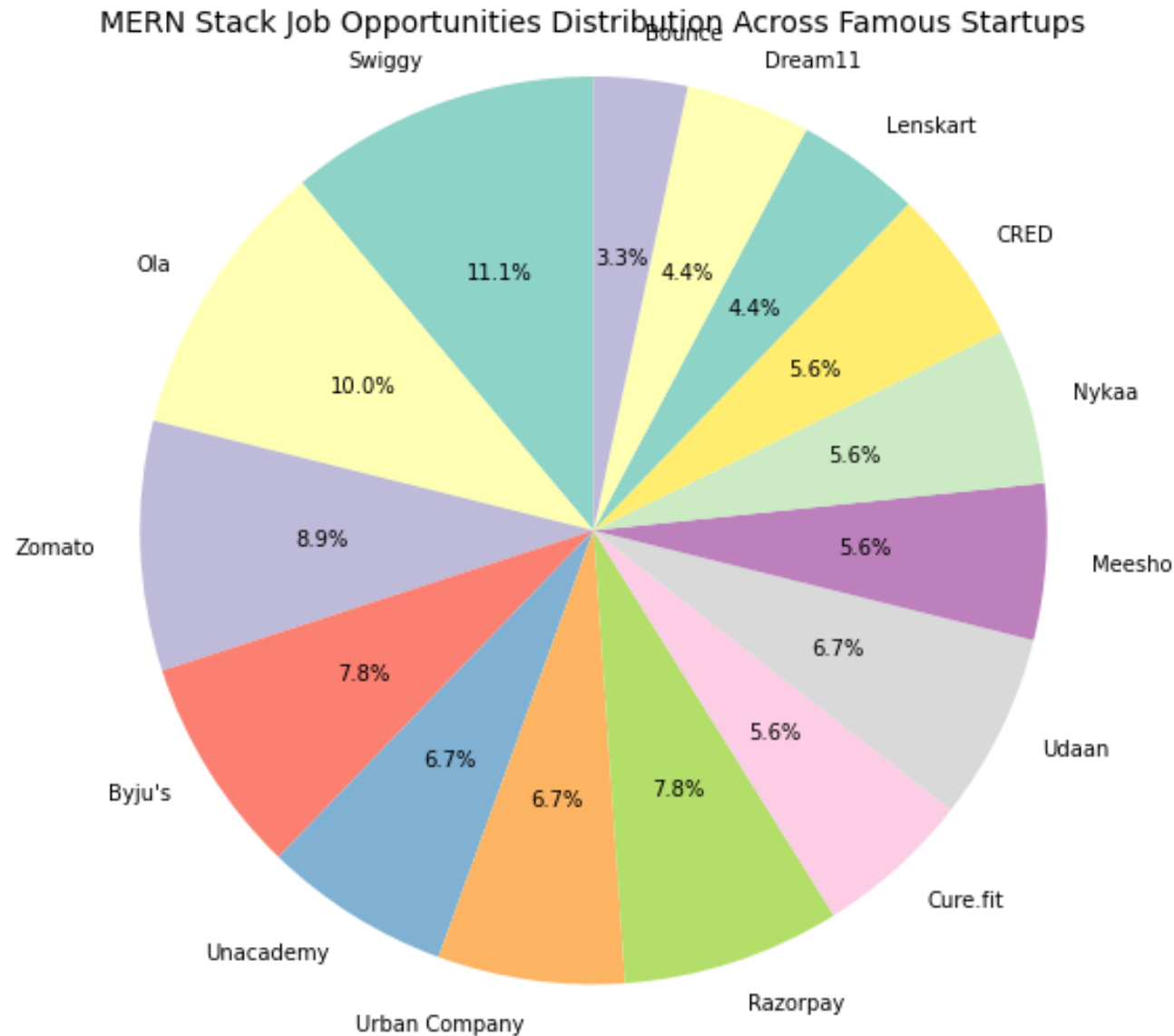


Full Stack Developer (Company wise job Trends)

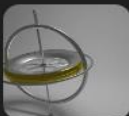
MERN Stack Job Opportunities Distribution Across Famous Companies



Full Stack Developer (Startups wise job Trends)



Popular apps built with React

	Facebook	▼		Instagram	▼		Discord	▼
	Wix	▼		Bloomberg	▼		Pinterest	▼
	Walmart	▼		SoundCloud Pulse	▼		Uber Eats	▼
	Airbnb	▼		Gyroscope	▼		Skype	▼
	Dropbox	▼		Netflix	▼		Tesla	▼
	WhatsApp	▼		Townске	▼		Delivery.com	▼
	Khan Academy	▼		Salesforce	▼		Instagram and react native	▼
	Myntra - Fashion Shoppin...	▼		Codecademy	▼		New York Times	▼

Data from Career Planning & Development :

CCPD II Openings with Blackrock



Sanjeev Srivastava

...

To: B.Tech 2024 Batch (for Placement); MCA 2024 Batch (For Placement)

Thu 14/03/2024 15:22

Cc: Sachin Kumar Goel; Corprelations ABESEC; Dean Training & Placement; Academic Heads; Dept. Placement Coordinators

Dear All,

Blackrock is hiring associate - java full stack

Qualification: B. Tech. & MCA.

Experience: 0-3 years

Salary: **11.00 LPA – 16.00 LPA**

Closing date for applications: **25-03-2024**

Apply Link: <https://careers.blackrock.com/job/20052759/associate-java-full-stack-gurgaon-in/>

All the best.

With warm regards,

Data from Career Planning & Development :

CCPD II Opportunities with various companies



Sanjeev Srivastava

...

To: B.Tech 2023 Batch (for Placement); MCA 2023 Batch (for Placement)

Fri 10/06/2022 15:02

Cc: Sachin Kumar Goel; Corprelations ABESEC; Academic Heads; Dept. Placement Coordinators

Dear All,

LimeChat is hiring Full Stack Intern for Bangalore. Candidates from B. Tech. background - 2023 batch are eligible for the LimeChat Summer Internship. Interns will be part of a cross-functional team that's responsible for the full software development life cycle, from conception to deployment. You may get the opportunity to work on existing, mission critical products, or build new ones from scratch. Details and Registration link are given below:

<https://limechat.zohorecruit.in/jobs/Careers/6075500000466107/Full-Stack-Intern?>

Microsoft Azure Hackathon

Eligibility: 2023/2024/2025 passouts females.

Link: <https://www.techgig.com/codegladiators?>

Data from Career Planning & Development :

CCPD II Baker Hughes is hiring Staff Full Stack Java Developers @ INR 7-8 LPA



Sanjeev Srivastava

...

To: B.Tech 2023 Batch (for Placement); MCA 2023 Batch (for Placement)

Mon 02/01/2023 17:11

Cc: Sachin Kumar Goel; Corprelations ABESEC; Academic Heads; Dept. Placement Coordinators

Dear All,

Kindly get registered ASAP to avoid the last-hour rush.

With warm regards,

Sanjeev Srivastava | AGM (Corporate Relations)

ABES Engineering College, Ghaziabad

Campus 1, 19th KM Stone, NH 24, Ghaziabad - 201 009, U.P.

Office: +91 120 7135112 (Extn.: **429**) | Fax: +91 120 7135115

W: <http://abes.ac.in>, E: sanjeev@abes.ac.in, M: **+91-9953320290**

LinkedIn Profile: <https://www.linkedin.com/in/sanjeevsvrivastava21/>

Data from Career Planning & Development :

CCPD II Convrse.ai is hiring 2023 Graduates



Sanjeev Srivastava

...

To: B.Tech 2023 Batch (for Placement); MCA 2023 Batch (for Placement)

Fri 03/03/2023 12:39

Cc: Corprelations ABESEC; Dean Training & Placement; Prabhat Singh; Academic Heads; Dept. Placement Coordinators

Dear All,

Convrse.ai, a technology company, is looking for Fullstack-Developer Intern and will pay **Rs. 20,000 per month** as a stipend. Post internship, based on performance, the CTC offered will be **INR 4.2 Lacs Per Annum**, latest by closing hours of **5th March 2023**.

Registration Link: https://skillenza.com/challenge/convrse-ai-fsd-intern-hiring-challenge-march-2023?utm_source=skillenza&utm_medium=email&utm_campaign=convrse-ai&utm_content=skillenza

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skillenza.com

Kindly apply ASAP before the link expires.

Data from Career Planning & Development :

Re: CCPD || Free AWS & Java Full Stack Program and placement Assistance by Tata STRIVE || CS Allied and MCA



Vivek Kumar Singh

...

To: B.Tech 2024 Batch (for Placement); MCA 2024 Batch (For Placement)

Fri 24/05/2024 14:57

Cc: Sachin Kumar Goel; Dean Training & Placement; Director; Dept. Placement Coordinators; Corprelations ABESEC

Dear All,
Kindly join WhatsApp group on below link

Dear Sir,
I request you to please send the link to all the unplaced candidates for 2022, 2023 and 2024.
Please find the link below:-
<https://chat.whatsapp.com/ILc5B3wH2J43ZZI6jFv3nq>

As per our discussion I request you to provide the approval for a whatsapp group by the name of ABES EC Warriors to share details of our programs with students and I am sharing the Group link with you as well so that you can get the access of the same group for the transparency between Tata and ABESEC.
I believe this step is going to be a revolutionary step for the Skill Development of INDIA by Tata STRIVE with the help of ABESEC.
I request you to kindly circulate the group link among the Final Semester, 2022 and 2023 passout students.
Note- We are going to Organise a Webinar on Google Meet so that we can counsel them for our courses and EGA test, Date - 27th may, Time - 12PM

Data from Career Planning & Development :

Happieloop Technologies is hiring for the role of Full Stack Developer Intern!



Vivek Kumar Singh

To: B.TECH. 2025 PLACEMENT

Cc: Corprelations ABESEC

...

Thu 28/11/2024 12:25

Responsibilities of the Intern:

- Develop and maintain websites, web applications, and other web-based projects
- Design and implement user interfaces and navigation structures
- Develop and maintain databases and content management systems
- Troubleshoot and debug web applications
- Optimize web applications for maximum speed and scalability
- Ensure cross-browser compatibility
- Collaborate with other developers and designers to create innovative solutions
- Stay up-to-date on emerging technologies

Data from Career Planning & Development :

CCPD II Opportunities with various companies



Sanjeev Srivastava

...

To: B.Tech 2023 Batch (for Placement); MCA 2023 Batch (for Placement)

Fri 10/06/2022 15:02

Cc: Sachin Kumar Goel; Corprelations ABESEC; Academic Heads; Dept. Placement Coordinators

Dear All,

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Eligibility: 2023/2024/2025 passouts females.

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Data from Career Planning & Development :

HOMIE s hiring for the role of Full Stack Developer!



Vivek Kumar Singh

...

To: B.TECH. 2025 PLACEMENT

Mon 30/12/2024 15:52

Cc: Corprelations ABESEC

Job Description

HOMIE s hiring for the role of Full Stack Developer!

Role Overview:

We're looking for a passionate Full-Stack Web Developer to take the lead in building and scaling HOM!E's web platform. You'll design our website, develop the brand dashboard, create analytics backends, and integrate APIs to bring our vision to life. This is a high-impact role with the opportunity to grow into a lead tech position as we scale.

Key Responsibilities:

- Develop and maintain HOM!E's web platform and analytics dashboard.
- Design secure and scalable backend systems for data processing and visualization.
- Integrate APIs to streamline social media and third-party platform connections.
- Collaborate with design and product teams to deliver a seamless user experience.

Git is important, and why it's important

What is Git?

- **Git is a version control system (VCS).**
- It helps developers **track changes** in their code, **collaborate with teams**, and **manage project history**.
- Originally created by *Linus Torvalds* (the creator of Linux kernel) in 2005.

Why is Git Important?

◆ 1. Version Tracking

- Every change is recorded.
- You can roll back to previous versions if something breaks.
- Prevents loss of work.

◆ 2. Collaboration

- Multiple developers can work on the **same project at the same time**.
- Git merges everyone's contributions smoothly.
- Avoids overwriting code.

◆ 3. Branching and Experimentation

- You can create a **branch** (a separate workspace) to test features/ideas.
- If it works → merge into main project.
- If not → discard without affecting the main code.

◆ 4. Backup & Remote Repositories

- With Git platforms (GitHub, GitLab, Bitbucket), your code is stored remotely.
- Provides backup and access from anywhere.

◆ 5. Accountability

- Each commit (change) records **who made it, when, and why**.
- Makes tracking bugs and responsibilities easier.

◆ 6. Industry Standard

- Almost all software companies expect developers to know Git.
- It's essential for open-source contributions, teamwork, and DevOps workflows.

<https://www.youtube.com/watch?v=OHcrb1XroMg>

Thank you